Adjoint mappings 1 : the power set

Adjoint mappings

Based on a note by Antoine Chambert-Loir and ideas of Patrick Massot.

For a set X, denote by $\mathcal{P}(X)$ the set of subsets of X and view it as an ordered set.

For $A \in X$, let $C_X(A)$ be the complement of A in X.

The mapping $C_X : \mathcal{P}(X) \to \mathcal{P}(X)^0$ is an isomorphism of $\mathcal{P}(X)$ with its opposite ordered set $\mathcal{P}(X)^0$.

The reciprocal isomorphism is C_X itself.

Adjoint mappings 2 : f lower star

Adjoint mappings (2)

For a mapping $f : X \to Y$,

define $f_* : \wp(X) \to \wp(Y)$

by

$$f_*(A) = \{ f(x) \mid x \in A \}.$$

($b \in f_*(A) \iff$ some f-premage of b belongs to A)

The mapping f_* is an increasing mapping in the sense that

$$A \subset A' \implies f_*(A) \subset f_*(A').$$

We have

$$f_*\left(\bigcup_{\lambda\in L} A_\lambda\right) = \bigcup_{\lambda\in L} f_*(A_\lambda).$$

For intersections, the inclusion

$$f_*\left(\bigcap_{\lambda\in L} A_\lambda\right) \subset \bigcap_{\lambda\in L} f_*(A_\lambda)$$

may fail to be an equality:



We have $f_*(\{0\}) = \{a\}$

$f_*(\{1\}) = \{a\}$

$f_*(\underbrace{\{0\} \cap \{1\}}_{=\emptyset}) = \emptyset$

whereas $f_*(\{0\}) \cap f_*(\{1\}) = \{a\}$

so the inclusion is strict.

Do we have $(g \circ f)_* = g_* \circ f_*$?

Adjoint mappings ③

For a mapping $f : X \longrightarrow Y$, define

$$f^* : \wp(Y) \longrightarrow \wp(X)$$

by

$$f^*(B) = \{x \in X \mid f(x) \in B\}.$$

The mapping f^* is an increasing mapping in the sense that

$$B \subset B' \Longrightarrow f^*(B) \subset f^*(B').$$

. The following square commutes:

$$\begin{array}{ccc} \wp(Y) & \xrightarrow{C_Y} & \wp(Y) \\ {\scriptstyle f^*}\downarrow & & \downarrow{\scriptstyle f^*} \\ \wp(X) & \xrightarrow{C_X} & \wp(X) \end{array}$$

In other words,

$$C_X f^*(B) = f^* C_Y(B)$$

for every $B \in \mathcal{P}(X)$.

We have

$$f^*\left(\bigcup_{\lambda \in L} B_\lambda\right) = \bigcup_{\lambda \in L} f^*(B_\lambda)$$

as well as

$$f^*\left(\bigcap_{\lambda \in L} B_\lambda\right) = \bigcap_{\lambda \in L} f^*(B_\lambda).$$

We have $A \subset f^* f_*(A)$

for every $A \in P(X)$, with equality if and only if $f^*(\{f(x)\}) \subset A$ for every $x \in A$.

Indeed, if $x \in A$, then

$$f(x) \in f_*(A),$$

so $x \in f^* f_*(A)$.

The inclusion is <u>strict</u> for

$$f : \{0,1\} \longrightarrow \{a\}$$

and $A = \{0\}$. We have

$$f^* f_*(A) = \{0,1\}.$$

We have $f_* f^*(B) \subset B$.

for every $B \in \mathcal{P}(Y)$, with equality if and only if $B \subset f_*(X)$.

Proof: The inclusion $f_* f^*(B) \subset B$ is obvious. In gory detail:

Let $y \in Y$. If

$y \in f_* f^*(B)$, then there exists $x \in f^*(B)$ such that $y = f(x)$. But $x \in f^*(B)$ implies that $f(x) \in B$. So $y \in B$.

If $B \subset f_*(X)$, we have

implies ...

It is also obvious that we have equality if $B \subset f_*(X)$.

Let us show that the inclusion

$$f_* f^*(B) \subset B$$

is strict if $B \not\subset f_*(X)$.

This is also clear:

if $B \not\subset f_*(X)$, then there exists $b \in B$ such that $b \neq f(x)$ for any $x \in X$.

In particular, $b \neq f(x)$ for any $x \in f^*(B)$. So the inclusion $f_* f^*(B) \subset B$ is strict. $\square$

Do we have

$$(g \circ f)^* = f^* \circ g^* \quad ?$$

(4)

Adjoint mappings : the first adjunction

Let $f : X \to Y$ be a mapping.

For subsets $A \subset X$ and $B \subset Y$, we have

$$f_*(A) \subset B \iff A \subset f^*(B).$$

Indeed, both sides amount to

"$f(a) \in B$ for every $a \in A$".

This property says that

f_* is a left adjoint of f^*

and f^* is a right adjoint of f_*.

Adjoint mappings 5 : f lower shriek

(5)

Adjoint mappings : $f_!$

For a mapping $f : X \longrightarrow Y$,

define $f_! : \mathcal{P}(X) \longrightarrow \mathcal{P}(Y)$

by

$$y \in f_!(A) \Leftrightarrow f^*(\{y\}) \subset A$$

(every f-preimage of y belongs to A)

(where $y \in Y$ and $A \in \mathcal{P}(X)$).

One can check that

$$f_!(A) = C_Y f_* C_X(A).$$

Proof: To prove the inclusion $\subseteq$, let $y \in f_! C_X(A)$; we have to show that $y \notin f_* C_X(A)$. In other words, we have to show that there is no $x \in C_X(A)$ such that $f(x) = y$. This is indeed the case by the definition of $f_!$: if $f(x) = y$, then $x \in A$, so $x \notin C_X(A)$.

Conversely, suppose that

$$y \in C_Y f_* C_X(A).$$

Then $y \notin f_* C_X(A)$.

So there is no $x \in C_X(A)$ such that $f(x) = y$. In other words, every $x \in X$ such that $f(x) = y$ is in A. This means that $y \in f_! C_X(A)$. $\square$

Note that $\tilde{f_!}$ is an increasing function:

$$A \subset A' \Rightarrow f_!(A) \subset f_!(A')$$

for all $A, A' \in \mathcal{S}(X)$.

$$E \xrightarrow{f} F \xrightarrow{g} G$$

Do we have $(g \circ f)_! = g_! \circ f_!$?

Yes indeed (so we get a functor

$$\mathcal{E} \longrightarrow \mathcal{E}$$

$$E \longmapsto \mathcal{P}(E)$$

$$f : E \to F \longmapsto f_! : \mathcal{P}(E) \to \mathcal{P}(F)$$

Let $A \in \mathcal{P}(E)$.

We have

$$(g \circ f)_!(A) = \{ \gamma \in G \mid (g \circ f)^*(\{\gamma\}) \subset A \}$$

$$\mathcal{P}(G) \xrightarrow{g^*} \mathcal{P}(F) \xrightarrow{f^*} \mathcal{P}(E)$$

$$(g \circ f)^* = f^* \circ g^*$$

$$= \{ \gamma \in G \mid f^* g^* \{\gamma\} \subset A \}.$$

Also

$$g_!(f_!(A)) = \{ z \in G \mid g^*\{z\} \subset f_!(A) \}$$

But $f_!(A) = \{ y \in F \mid f^*\{y\} \subset A \}$.

Therefore

$$g_!(f_!(A)) = \{ z \in G \mid f^* g^*\{z\} \subset A \}.$$

$$= (g \circ f)_!(A)$$

Adjoint mappings 6 : the second adjunction

The second adjunction (6)

Let $f : X \to Y$ be a mapping.

We have seen that

$$f_* : \mathcal{P}(X) \longrightarrow \mathcal{P}(Y)$$

is a left adjoint of

$$f^* : \mathcal{P}(Y) \longrightarrow \mathcal{P}(X)$$

in the sense that

$$f_*(A) \subset B \iff A \subset f^*(B)$$

for every $A \in \mathcal{P}(X)$ and $B \in \mathcal{P}(Y)$.

It so happens that f_* is a right adjoint of f^*.

We have to prove

$$f^*(B) \subset A \iff B \subset f_!(A)$$

Proof of $f^*(B) \subset A$

$\iff B \subset f_!(A)$.

for $A \in \mathcal{P}(X)$, $B \in \mathcal{P}(Y)$.

Suppose $f^*(B) \subset A$ and let $y \in B$. Then $\{y\} \subset B$,

so $f^*(\{y\}) \subset f^*(B)$,

so $f^*(\{y\}) \subset A$,

so $y \in f_!(A)$.

This proves that $B \subset f_!(A)$.

Conversely, suppose that

$B \subset f_!(A)$ and let $x \in f^*(B)$;

we have to show that $x \in A$.

Now, $x \in f^*(B)$

$\Rightarrow f(x) \in B$

$\Rightarrow f(x) \in f_!(A)$

$\Rightarrow x \in A.$

□